

August 31, 2026

Editor
Physics of Fluids

Dear Editor,

Please consider our substantially revised manuscript, “Bulk-to-Wall Observability in Rarefied Hypersonic Flow over Triangular Protrusions: Full-Range Moment Limits and Incident Half-Range Sufficiency,” by Elyas Lekzian and Ehsan Roohi, for publication in *Physics of Fluids*.

The earlier version centered on a model-conditional spatial-support diagnostic, R_{95} , extracted from direct simulation Monte Carlo (DSMC) macroscopic fields. The review correctly identified that this evidence was useful but insufficient for a general physical claim about wall-load observability, particularly for shear. We have therefore retained the full validation and reproducibility structure while rebuilding the scientific center of the manuscript around new DSMC data and a kinetic sufficiency test.

The revised paper makes three main contributions. First, an equal-capacity blind interpolation experiment compares primitive variables with the complete momentum-flux tensor and a heat-flux augmentation. The stress tensor improves pressure relative to primitive fields, but finite full-range states do not improve transferable signed shear. Second, a constructive positive-distribution example proves that full-range moments through degree three can be identical even though incident half-range pressure differs and half-range shear changes sign. Third, collision tallies from the Stochastic PARallel Rarefied-gas Time-accurate Analyzer (SPARTA) show that the incoming half-range state, combined with the known diffuse-wall kernel, reconstructs co-temporal DSMC pressure with 0.15–0.28% normalized root-mean-square error (NRMSE) and signed shear with 1.30–7.35% NRMSE across six geometry–rarefaction cases. Tests against independent 40,000-step windows retain 1.68–2.85% pressure error. They also identify the forward-facing shear cases as sampling-noise limited, so those cases are reported explicitly rather than concealed behind absolute shear.

The significance of this result extends beyond one surrogate architecture. A kinetic–continuum interface or learned wall closure can fail even when it receives accurate low-order macroscopic fields because full-range moments combine molecules approaching and leaving the wall. The incident half-range result provides a physically interpretable upper-bound state for designing practical directional-moment or reduced-distribution closures. The neural surrogate remains in the paper as a validated emulator and controlled comparison tool; it no longer carries the central kinetic claim. Likewise, R_{95} is retained only as a secondary model-conditional diagnostic.

The manuscript is a strong fit for *Physics of Fluids* because it connects rarefied-gas dynamics and gas–surface interaction with uncertainty-aware surface loading and data-driven closure assessment. All figures have been rebuilt in publication-quality landscape form. The reproducibility package records the cases and SPARTA inputs together with block-resolved moment and collision outputs; it also supplies the exact commands, packed arrays, numerical quality-control reports, source scripts, and checksums. Publication graphics are included in both Portable Document Format (PDF) and Scalable Vector Graphics (SVG) formats.

This manuscript is original, is not under consideration elsewhere, and has been approved by both authors. The authors declare no conflict of interest and no external funding for this work.

Thank you for your consideration.

Sincerely,

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